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Comparisons of soil-water content between a Moso bamboo (*Phyllostachys pubescens*) forest and an evergreen broadleaved forest in western Japan

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Abstract

In Japan, forests of Moso bamboo (*Phyllostachys pubescens*, an exotic invasive giant bamboo) have naturalized and expanded rapidly, replacing surrounding broadleaved and coniferous forests. To evaluate impacts caused by these forest-type replacements on the hydrological cycle, soil-water content and its spatial variability in a Moso bamboo forest were compared with those in an adjacent evergreen broadleaved forest, in a case study of a stand in western Japan (northern Kyushu). The volumetric soil-water content averaged over depths between 0 and 60 cm was consistently higher in the bamboo stand than that in the broadleaved stand. These results contrast with previous studies comparing the soil-water content in Moso bamboo forests with that in other forest types. The sum of canopy transpiration and soil evaporation (*E*) in the bamboo stand tended to be larger than that in the broadleaved stand. Small canopy interception loss was reported in the bamboo forest. Therefore, the large amount of *E* would counterbalance the small canopy interception loss in the bamboo forest. Differences in soil characteristics between the two stands may be the main factor causing differences in soil-water content. Spatial variation in soil-water content in the bamboo stand was larger than that in the broadleaved stand, confirming findings in a previous series of our study. This could happen because the well-developed root-system in the bamboo forest enhances preferential flow in the soil. To evaluate the effects of aggressive invasion of alien giant bamboo on the ecosystem functions, we recommend further studies measuring various hydrological components in various Moso bamboo forests.

Keywords: bamboo forest, canopy transpiration, soil moisture, soil water budget method, spatial variability.

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Introduction

Moso bamboo (*Phyllostachys pubescens*, a giant bamboo of Poaceae) is a main forest type in eastern Asian countries (e.g., Chen *et al*. 2009; Chung *et al*. 2009). Moso bamboo was introduced from China to Japan in the 1700s and has been widely used for various purposes such as timber and food. Recently, however, abandoned Moso bamboo forests have become more widespread because of a reduction in bamboo usage and an increase in bamboo sprout imports. Since the vegetative reproduction (rhizomatous clonal growth) of Moso bamboo is aggressive, Moso bamboo forests have naturalized and expanded rapidly, and are replacing surrounding broadleaved and coniferous forests (Okutomi *et al*. 1996; Nishikawa *et al*. 2005; Suzuki & Nakagoshi 2008; Suzuki 2015). Drastic changes not only in biodiversity but also in ecosystem functions could be expected in the processes of such expansions and shifts in forest types (cf. Kobayashi & Tada 2010; Fukushima *et al*. 2015; Umemura & Takenaka 2015).

The hydrological cycle in forests is strongly affected by changes in vegetation, that is, species composition and community structure of plants (Swank & Douglass 1974; Bosch & Hewlett 1982). Some researchers have speculated that the rapid expansion of Moso bamboo forests could result in changes in the hydrological cycle that may increase the frequency of shallow landslides, soil erosion, floods, and water shortages (e.g., Ishiga *et al*. 2001; Hiura

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et al. 2004). More than half the prefectures of Japan have introduced local taxes related to forest management (e.g., Imawaka & Sato 2008), and the primary purpose of these taxes is to fund soil and water conservation in degraded forests, including the financing of efforts to manage the above problems. Therefore, some prefectures use tax revenues to replace Moso bamboo forests with other forest types.

Observations of various hydrological components are required in various types of forest to evaluate and compare the differences in frequencies of shallow landslides, soil erosion, floods, and water shortages among different forest types. Soil-water content is a fundamental hydrological component in forests. The above-mentioned problems are directly or indirectly related to soil-water condition. The occurrence of shallow landslides and the magnitude of peak flows during heavy rainfall events are generally affected by soil water conditions prior to heavy rainfall (Gray & Sotir 1996; Tsukamoto 1998). Canopy transpiration (Kume *et al*. 2007; Wang *et al*. 2010), which is the dominant component of evapotranspiration and strongly affects river-flow regimes (Brown *et al*. 2005; Komatsu *et al*. 2007, 2011), interacts soil-water content. Liang *et al*. (2010) pointed out slope stability changes with spatial variations in soil-water content.

However, few studies have compared soil-water content in Moso bamboo forests with that in other forest types. Torii (2007) reported that Moso bamboo forests have lower surface soil-water content than Japanese cedar (*Cryptomeria japonica*), Japanese cypress (*Chamaecyparis obtusa* Endlicher), deciduous broadleaved, and orchard forests in Kochi, Japan. Yokoo *et al*. (2005) compared soil-water content of three forest types and found Moso bamboo forest had lower soil-water content than a Japanese cypress forest and a mixed forest in Kumamoto, Japan. At this stage, we do not know whether soil-water content in Moso bamboo forests is necessarily lower than that in other forest types.

In the present study, we continuously measured soil-water content at three depths at several points in a stand of bamboo forest and a stand of an adjacent evergreen broadleaved forest in Fukuoka, Japan for 11 months. Soil volumetric water content and its spatial variability in the bamboo stand were compared with that in the broadleaved stand. In addition, we calculated the sum of canopy transpiration and soil evaporation in the two different types of stand using the measured soil-water content data. In this bamboo stand, we previously reported observations of various hydrological components such as canopy transpiration, canopy interception loss, overland flow, and biomat flow (Kume *et al*. 2010; Komatsu *et al*. 2010b; Kuramoto *et al*. 2011; Shinohara *et al*. 2013). Biomat flow is the movement of stormwater through a near-surface layer on hillsides (Sidle *et al*. 2007). Ide *et al*. (2010) also reported differences in soil characteristics between the bamboo and broadleaved stands analyzed in the present study. Using their findings, we tried to identify the main factor producing differences in soil-water content between the Moso bamboo stand and broadleaved stand.

Material and methods

Site description

Experiments were carried out in a Moso bamboo stand and an adjacent evergreen broadleaved stand in the Kaguya site in the Kasuya Research Forest of Kyushu University, 12 km east of the city of Fukuoka, Japan (33°37′N, 130°31′E, 80 m a.s.l.). The underlying bedrock is sandstone including lime, classified as being part of the Sangun metamorphic belt. Soils are more than 60 cm deep. The mean air temperature recorded between 1995 and 2005 at the nearest meteorological observatory in Fukuoka was about 16°C, and the mean annual precipitation was 1790 mm. The bamboo stand at the study site was about 70 m from north to south and about 50 m from east to west. An evergreen broadleaved stand surrounds the bamboo stand and both stands face west and have a slope of less than 10°.

The bamboo forest forms a 15 m tall continuous canopy (overstory) layer and the mean diameter at breast height (DBH) of the stems is about 11 cm. The bamboo forest was thinned in April–May of 2007, after which the bamboo density was about 4000 bamboo stems ha^{-1}. This closed canopy forest has a mean leaf-area index of 2.4 m^2/m^2, measured using a digital nonspherical color photograph (Nikon COOLPIX990, Tokyo, Japan) and Gap Light Analyzer software (Frazer *et al*. 1999). The forest floor had scarce vegetation cover. *Neolitsea sericea* (Blume) Koidz. and *Cinnamomum camphora* (L.) Presl., which are species of the warm-temperate region of Japan, dominated the broadleaved forest with a mean DBH of 13.1 cm and 23.6 cm, respectively. The height of the canopy layer in the broadleaved forest is comparable to the canopy height of the bamboo forest. Moso bamboo invaded the broadleaved forest from the nearby bamboo forest. The bamboo density of the broadleaved forest is about 500 stems/ha and the mean DBH of the bamboo growing there is similar to the mean DBH of the bamboo forest.

Measurements

Ide *et al*. (2010) selected a 6 m × 6 m plot in each of the bamboo and broadleaved stands. Volumetric soil-water content was measured at depths of 10, 30, and 50 cm at five points in the bamboo stand and four points in the broadleaved stand using a dielectric probe sensor (EC-10, Decagon Device Inc., Pullman, WA). These points were set up around the 6 m × 6 m plot in both stands. Each signal cable was connected to a data logger with a peripheral multiplexer (CR1000 and AM16/32, Campbell Scientific Inc.) and data were collected at 10-min intervals. Soil-water content was calculated using the general relationship between output voltage of the EC-10 sensor and soil-water content. We assumed that soil-water content at depths of 10, 30, and 50 cm represented soil-water content at depths of 0–20 cm, 20–40 cm, and 40–60 cm depths, respectively. Soil-water content in the 0 and 60 cm range ($\theta_{\_0\text{–}60}$) was calculated using the average soil-water content at the three depths. $\theta_{\_0\text{–}60}$ data were then used to calculate the average soil-water content ($\theta_{\_ave}$) in each stand. Some data were missing during the experimental period at two points in both stands. The $\theta_{\_ave}$ vale was calculated using $\theta_{\_0\text{–}60}$ at three points for the bamboo stand and at two points for the broadleaved stand. The soil-water content measurements were conducted between July 2008 and May 2009. During the experimental period, precipitation was measured at an open site about 300 m west of the study area using a tipping-bucket rain gauge with a resolution of 0.5 mm (Ohtakeiki, RA-1, Tokyo, Japan). The tipping-bucket rain gauge was connected to a data logger (HOBO Event, Onset Computer, Bourne, MA) and the tipping time was recorded.

Soil-water budget method

Assuming no developed understory layer was present in both forests, evapotranspiration in forests can be divided into canopy transpiration, canopy interception loss, and soil evaporation. Canopy transpiration and canopy interception loss are much larger than soil evaporation in forests of Japan. We calculated the sum of canopy transpiration and soil evaporation (E) using $\theta_{\_ave}$ and the soil-water budget method (e.g., Wilson *et al*. 2001) in both stands. For a given day, E was calculated as the difference between $\theta_{\_ave}$ on the previous day and $\theta_{\_ave}$ on the current day. In this calculation, we applied the following two assumptions. First, water at depths greater than 60 cm below the soil surface did not contribute to canopy transpiration and soil evaporation. Second, water did not move between the soil above and below 60 cm deep. Since the second assumption would not apply to periods during rainfall and shortly afterward, we eliminated all days with rainfall and up to two additional days after rainfall. Consequently, E was obtained for 69 days during the experimental period.

In the bamboo forest, canopy transpiration was measured using the sap-flow technique (Onozawa *et al*. 2009b; Kume *et al*. 2010; Komatsu *et al*. 2010b). The value of E in the bamboo stand was compared with the canopy transpiration based on the sap-flow technique. Canopy transpiration measurements were conducted in a 10 × 10 m plot close to the points where soil-water content was measured. Stand-scale transpiration (E_{sap}) was calculated as: $E_{sap} = J_S \times A_{Sstand}/A_G$, where E_{sap} (m/s) is the stand-scale transpiration based on the sap-flow technique, J_S ($m^3/m^2/s^1$) is the mean stand sap flux, A_{Sstand} (m^2) is the stand culm sectional area, and A_G is the surface area (= 10 m^2). A_{Sstand} is the sum of the culm sectional area for each bamboo stem (A_{Sb}) (m^2). A_{Sb} was obtained from the DBH and culm thickness data. The J_S value was calculated using A_{Sb} and the sap flux density (F_d) ($m^3/m^2/s^1$) for each bamboo stem. The value of F_d was measured for 16 individuals using the Granier-type sensor between November 2007 and February 2009. Kume *et al*. (2010) provides a complete description of transpiration measurements.

Results

The soil-water content at each depth in both stands varied with the temporal variation in precipitation (Fig. 1a,b,c). A continuous decrease in the soil-water content was observed in the two periods (i.e., in July 2008 and in November 2008) in both stands because only a small amount of precipitation occurred. In the last phase of the two periods, the soil-water content at the depth of 10 cm ($\theta_{\_10}$) in the broadleaved stand was extremely low, approximately 0%, and $\theta_{\_10}$ in the bamboo stand was approximately 10%. Soil-water content at the three measured depths was different between the two stands. In the bamboo stand, soil-water content at the depth of 50 cm ($\theta_{\_50}$) was consistently larger than those at 10 and 30 cm deep ($\theta_{\_10}$ and $\theta_{\_30}$). In the broadleaved stand, $\theta_{\_30}$ was the largest, followed by $\theta_{\_50}$ and $\theta_{\_10}$. The $\theta_{\_ave}$ value in the bamboo stand was larger than in the broadleaved stand throughout the experiment (Fig. 1d). Although $\theta_{\_10}$ and $\theta_{\_50}$ values in the bamboo stand were larger than those in the broadleaved stand when averaged over the experimental period, $\theta_{\_30}$ averaged over the same time in the bamboo stand was smaller than in the broadleaved stand (Fig. 2a).

The difference between the maximum and minimum soil-water content averaged over the experimental period in the bamboo stand was larger than that in the broadleaved stand at any depth (Fig. 2a). Figure 2b compares soil-water content averaged over 6 days between July 5 and July 10, 2008 between the two stands. Data at the largest number of points were available for this time period. The difference between the maximum and minimum soil-water content averaged over these 6 days in the bamboo stand was also larger than that in the broadleaved stand at any depth.

Figure 3 compares temporal variations in E and E_{sap} in the bamboo stand. This figure is based on E for 69 days during which the above two assumptions apply (see

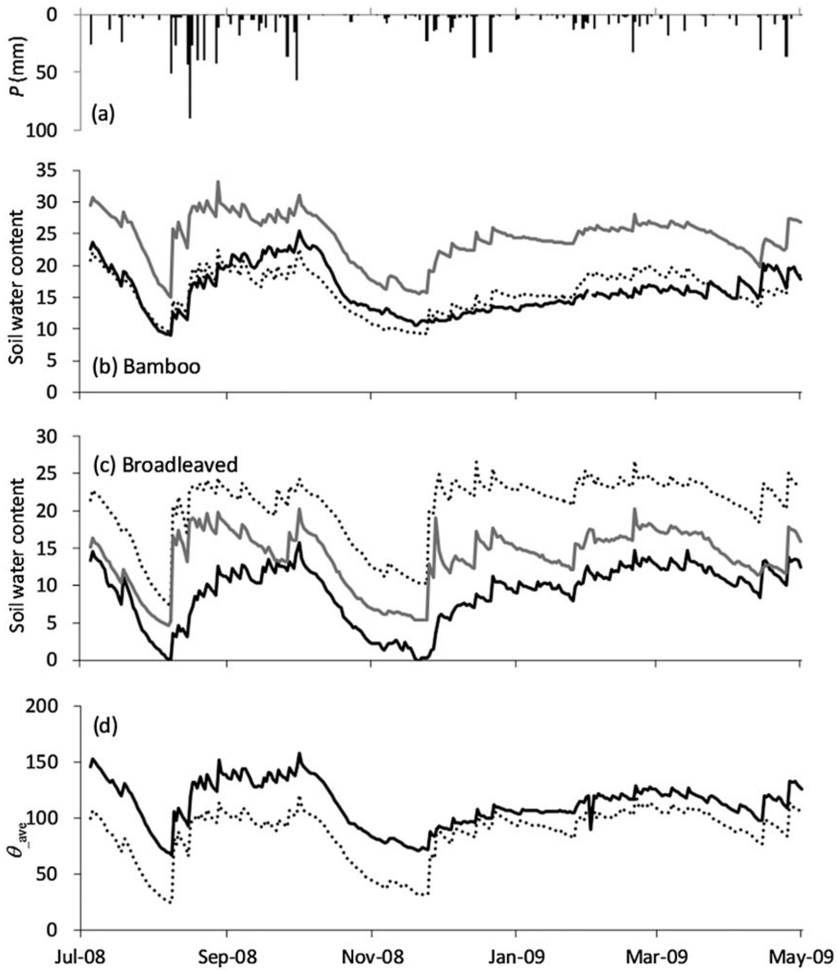

Fig. 1 Time series of (a) precipitation (P); (b, c) soil-water content at depths of 10, 30, and 50 cm in the bamboo stand and broadleaved stand, respectively; and (d) average soil-water content ($\theta_{\_ave}$) (mm) during the period July 5, 2008 to May 1, 2009. This figure relies on data at the three points in the bamboo stand and two points in the broadleaved stand. These five points did not include missing data. (b,c) ——, 10 cm; ······, 30 cm; ——, 50 cm. (d) ——, bamboo; ······, broadleaved.

above) and E_{sap} for all days except for days without measurement. Temporal variation in E was similar to that in E_{sap}. E was significantly correlated to E_{sap} ($P < 0.05$, $R = 0.81$). The value for E in the bamboo stand was significantly correlated to that in the broadleaved stand ($P < 0.01$, $R = 0.933$), although the value for E in the bamboo stand tended to be larger than that in the broadleaved stand (Fig. 4). The average E value in the bamboo stand (1.9 mm day^{-1}) was slightly larger than that in the broadleaved stand (1.8 mm day^{-1}). The value for E in the broadleaved stand was calculated as less than 0 for three of the days, which was probably caused by water input from fog drip, dynamic change in soil temperature, or sensor noise. Even when we eliminated data for those three days, the above results were not altered ($P < 0.01$, $R = 0.945$, slope = 1.03).

Discussion

Torii (2007) and Yokoo *et al*. (2005) previously reported that soil-water content in bamboo forests was lower than that in other forest types. In the present study, $\theta_{\_30}$ in the bamboo stand was lower than $\theta_{\_30}$ in the broadleaved stand. Both $\theta_{\_10}$ and $\theta_{\_50}$ in the bamboo stand were higher than $\theta_{\_10}$ and $\theta_{\_50}$ in the broadleaved stand, respectively. Consequently, $\theta_{\_ave}$ in the bamboo stand was higher than $\theta_{\_ave}$ in the broadleaved stand. These results suggest that soil-water content in bamboo forests is not necessarily lower than that in other forest types. This is consistent with Umemura & Takenaka (2015). They reported that soil-water contents in Moso bamboo forests were not clearly different from those in Japanese cypress and mixed (Moso bamboo and Japanese cypress) forests in three sites of central Japan. Yokoo *et al*. (2005) measured soil-water content in a Moso bamboo forest with an extremely high stem density (i.e., 13200 stem/ha), which is approximately three times higher than stem density in our Moso bamboo forest. Although Torii (2007) did not show stem density for three Moso bamboo forests measured in his study, differences in stem density might cause the different results between our study and the other two studies (Yokoo *et al*. 2005; Torii 2007).

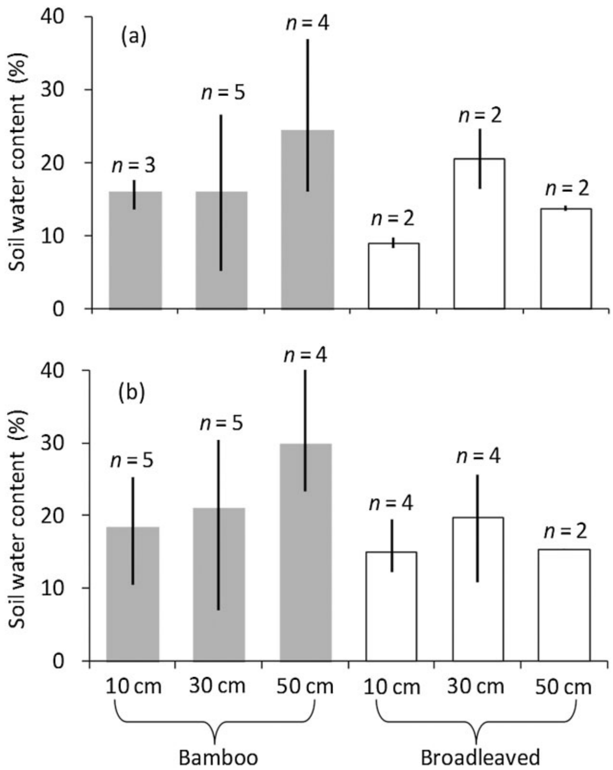

Fig. 2 Soil-water content at three depths in the bamboo and broadleaved stands during (a) the period July 5, 2008 to May 1, 2009 and (b) July 5, 2008 to July 10, 2008. Top and bottom of vertical bars indicate the maximum and minimum of each sample size (n), respectively.

Soil-water content measured using a dielectric probe sensor is affected by soil temperature and electrical conductivity (EC). When soil-water content is constant, output voltage increases with increasing soil temperature and EC (Saito *et al*. 2008; Miyamoto *et al*. 2009). That is, soil-water content tends to be overestimated when measured in very warm soil with high EC. To prevent the overestimates, soil-water content is generally calculated using a relationship between output voltage and soil-water content that is preliminarily obtained in each site. Although such calibration was not conducted in both of the forests, the above-mentioned results would not be altered because: (i) the two forests are adjacent, implying they have similar levels of soil temperature and EC; and (ii) Yokoo *et al*. (2005) reported that there were not clear differences in EC between a Moso bamboo forest and an adjacent coniferous forest. This implies that the EC in the bamboo forest would not be considerably larger or smaller than the EC in other forest types.

Torii (2007) supposed that low soil-water content in Moso bamboo forests could be caused by high canopy transpiration in Moso bamboo forests. Actually, E_{sap} in the Moso bamboo forest of our site was larger than E_{sap} in coniferous forests by 12% of annual precipitation (Komatsu *et al*. 2010b). In addition, E in the bamboo stand tended to be larger than that in the broadleaved stand at this site (Fig. 4). In general, canopy transpiration is considerably larger than soil evaporation in forests with closed canopies (Vertessy *et al*. 2001; Iida *et al*. 2006; Daikoku *et al*. 2008). Thus, canopy transpiration in the bamboo forest would be larger than that in the broadleaved forest.

Shinohara *et al*. (2013) observed canopy interception loss in the bamboo forest, and reported that the ratio of canopy interception loss to rainfall in the bamboo forest (= 9.2%) was considerably smaller than that in other forest types such as coniferous and broadleaved forests. A small canopy interception ratio was also observed in other Moso bamboo forests (Hattori & Abe 1989; Onozawa *et al*. 2009a). The sum of canopy interception loss and canopy transpiration in the bamboo forest was comparable to that in coniferous plantation forests because the large canopy transpiration in the bamboo forest compensates for the small canopy interception loss (Komatsu *et al*. 2010b). Systematically larger or smaller levels of evapotranspiration in broadleaved forests in comparison to coniferous forests have not been observed in Japan (Komatsu *et al*. 2008, 2010a). This suggests that the sum of canopy transpiration and canopy interception loss in a broadleaved forest could be comparable to those in Moso bamboo forests.

The infiltration rate in the bamboo forest was higher than that in the broadleaved forest (Ide *et al*. 2010). Kuramoto *et al*. (2011) measured overland flow at six plots in the same bamboo forest studied here and reported that the ratio of overland flow to rainfall ranged between 0.6% and 35.5%. The values were apparently lower than those observed in Japanese cypress forests and slightly lower than those observed in Japanese cedar and broadleaved forests. Further, overland-flow and biomat flow have both been considered to be important pathways of catchment runoff in recent years (Sidle *et al*. 2007). Hiura *et al*. (2004) pointed out that biomat flow might be an important process in Moso bamboo forests, based on their experiments using artificial rainfall. Kuramoto *et al*. (2011) measured biomat flow in the upper 10 cm of soil in three plots in the same bamboo forest studied here, and reported that only a small amount of biomat flow was observed in this bamboo forest. These small amounts of overland flow and biomat flow in this bamboo forest could increase water infiltration into the soil and therefore soil-water content.

Soil porosity in the bamboo forest was slightly higher than that in the broadleaved forest at any soil depth (Ide *et al*. 2010). Wang and Liu (1995) reported that effective water-retention capacity at soil depths of the 0–60 cm in a Moso bamboo forest was larger than that in a Chinese fir plantation and a broadleaved forest. As previously noted, high levels of canopy transpiration in the bamboo forest

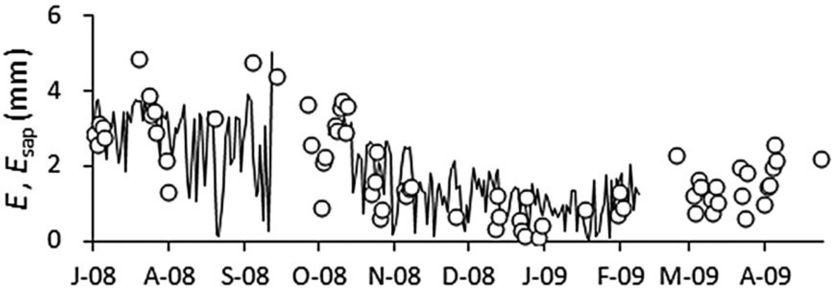

Fig. 3 Temporal variations in the sum of canopy transpiration and soil evaporation based on the water-budget approach (E) and canopy transpiration based on the sap-flow technique (E_{sap}) in the bamboo stand. Circles and line indicate E and E_{sap}, respectively.

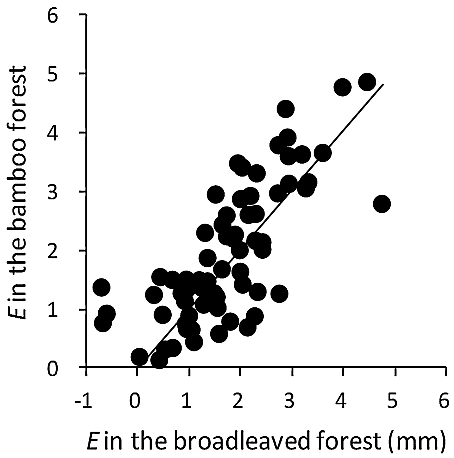

Fig. 4 Comparison between sum of canopy transpiration and soil evaporation (E) in the bamboo stand and E in the broadleaved stand. A regression line was determined using the least-squares method ($y = 1.01 \times x$).

may be counterbalanced by a small rate of canopy interception loss. Furthermore, in the present study, although the differences in θ_{ave} between the two forests studied in summer were larger than those in winter, the differences were nearly constant throughout the summer or winter (Fig. 1d). This implies that water balance (i.e., the differences between input and output water in the soil layers) was comparable between the two forest types. Thus, the difference in soil characteristics between the Moso forest and the broadleaved forest may be the main factor causing the differences in soil-water content between the two forests in the present study.

Spatial variation in the soil-water content in the bamboo stand was greater than that in the broadleaved stand at any soil depth (Fig. 2), which confirms Torii's (2007) observations. Moso bamboo forests have well-developed root systems (Kawai *et al*. 2008). Ide *et al*. (2010) reported fine root biomass in the bamboo stand was more than that in the broadleaved stand at the same site studied here. These root systems would enhance preferential flow in the soil (Johnson & Lehmann 2006; Liang *et al*. 2011), and increase spatial variation in soil-water content. However, note that water would not be carried into deeper soil layers (i.e., soil depth is > 30 cm) because the root system, especially the rhizomes, in Moso bamboo forests are well developed only in the shallow soil layers (i.e., soil depth is ≤ 30 cm) (Isagi *et al*. 1997).

Conclusions

We continuously measured soil-water content at three depths at several points in a Moso bamboo stand and a broadleaved stand for 11 months. The soil-water content and its spatial variability in the bamboo stand were compared with those in the broadleaved stand. The soil-water content in the bamboo forest was not lower than that in the broadleaved forest, which is different from previous studies (Yokoo *et al*. 2005; Torii 2007) that compared the soil-water content of Moso bamboo forests with those of other forest types. Spatial variation in soil-water content in the bamboo stand was greater than that in the broadleaved stand, which confirmed the finding of a previous study (Torii 2007). Since characteristics of Moso bamboo forests such as stem density and root amounts differ from one Moso bamboo forest to another, we recommend further studies measuring various hydrological components in various Moso bamboo forests, in order to understand the effects of aggressively invading alien giant bamboo on the ecosystem functions in Japanese forests.

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